INFRASTRUCTURE CAPEX ANALYSIS

Asset Valuations & Functional Mapping

Lwandile Engineering Projects (Pty) Ltd | Mpumalanga

Master Capital Allocation Matrix

Infrastructure Entry	CAPEX Value (ZAR)	Primary Strategic Function
30kW/30kWh Industrial BESS & Solar PV Grid	R 650,000	Thermal reserve stability & baseload buffer
Biogas Digester & Co-Gen Generator	R 420,000	Closed-loop electricity and hot steam generation
Stage 1: Swine Nursery Pod Structure (48 m^2)	R 280,000	Airtight physical pod pathogen isolation
Stage 2: Swine Finisher Barn (384 m^2)	R 450,000	High-occupancy biomass grow-out environment
Poultry Sprint Barn & Systems (384	R 550,000	Passive aerodynamic convective

Supporting Lifelines Valuation

Infrastructure Entry	CAPEX Value (ZAR)	Primary Strategic Function
Hydrology, Solar Pumping & Elevated Water Tower	R 180,000	Uninterrupted multi-day reserve water network
Slabs, Civil Ground Works & Gravity Drainage	R 220,000	Passive 2% gradient waste delivery (pump-free)
Biosecurity Gates, Wash Bays & Change Block	R 95,000	Shielding vehicle & human entry pathogens
IoT Sensor Network & Automation Control	R 75,000	Real-time microclimate and consumption alerts
Project Contingency & Professional Commissioning	R 180,000	Civil, electrical, and thermal integration sign-off

Solar & BESS Infrastructure Details

CAPEX: R 650,000

Ensures energy sovereignty for critical electrical components. Protects operations from external grid instability.

- **30kW Hybrid Inverter:** Manages high-inductive loads from water pumping and mechanical augers.
- **30kWh Lithium Iron Phosphate (LFP):** Provides deep-cycle storage reserves for nighttime lighting.
- **25kWp Solar Array:** Roof-mounted monocrystalline PV panels generate steady daily recharges.

Functional Role

Functions as the main energy shield, delivering consistent power to backup electric heating mats, automated feed lines, and data dashboard networks.

Biogas & Co-Gen Loop Details

CAPEX: R 420,000

Converts agricultural waste into free thermal and electrical power, securing extreme cold weather resilience.

- **60 m^3 Anaerobic Digester:** Processes organic animal manure slurry completely off-grid.
- **12kVA Gas Generator Set:** Converts captured methane directly into electrical volts.
- **Heat Exchanger Jacket:** Extracts exhaust engine heat to produce high-temperature steam lines.

Functional Role

Supplies direct subsurface floor steam heat to the swine nursery pods, removing massive heating thermal loads from the BESS batteries during cold highveld nights.

Swine Stage 1: Nursery Pods

CAPEX: R 280,000

Constructs a high-insulation physical boundary block to protect vulnerable, newly weaned piglets.

- **Insulated Cavity Walls:** Double-skin brick design locks inside microclimate heat perfectly.
- **Elevated Polypropylene Floor:** Soft plastic slatted grids keep animal beds dry and sanitary.
- **4 Sealed Chambers:** Vapor-tight doors isolate populations into secure 12 m^2 sub-units.

Functional Role

Provides immunological air-gapping, isolating weaners from older finishing hogs and capping structural pathogen spread to a maximum 25% exposure limit.

Swine Stage 2: Finisher Barn

CAPEX: R 450,000

Builds the heavy grow-out payload barn designed for large physical volumes and high metabolic body heat.

- **Open-Sided Portal Frame:** Structural steel frame maximizing passive cross-wind cooling.
- **Deep-Litter Composting Bed:** Composting sawdust layer generates insulation and organic manure fertilizer.
- **Winch Curtain Curbs:** Heavy PVC canvas shields hogs from high-wind storm fronts.

Functional Role

Houses up to 160 hogs simultaneously through their 16-week growth cycle, shifting metabolic output into heavy market-ready weight payload.

Poultry Sprint Barn Details

CAPEX: R 550,000

A specialized, high-velocity poultry barn built to optimize passive air flows and prevent thermal shock cycles.

- **Thermal Ridge Chimney:** 18-degree roof pitch generates passive siphon airflow naturally.
- **50mm PIR Ceiling Board:** Premium roof insulation blocks extreme radiant summer sun loads.
- **Internal Brooding Isolation:** Curtains seal off 128 m^2 for energy-efficient Day 1-14 heating.

Functional Role

Hosts 5,760 broilers per batch. Passive convective cooling eliminates the need for heavy electric extraction fans, protecting BESS reserve levels.

Civil Works & Gravity Slurry Channels

CAPEX: R 220,000

Structural site preparation and gravity-fall concrete engineering to ensure zero-power waste management.

- **Reinforced Slabs:** Load-bearing concrete foundations for heavy structural integrity.
- **2% Gradient Drainage Channels:** Concrete floor channels poured with precise downward slopes.
- **Primary Collection Sump:** Deep basin collecting gravity runoff at the lowest site point.

Functional Role

Leverages hydraulic head pressure to move manure slurry directly into the digester, completely eliminating high-wattage collection pumps and mechanics.

Hydrology & Water Security Systems

Solar Borehole Pumping

Low-voltage DC submersible pump pulls from deep aquifers during peak sun hours, drawing zero load from the BESS storage batteries.

Elevated Storage Tower

Gravity-fed water distribution block holding 10,000L. Secures 5 days of uninterrupted backup operational supply for absolute farm resilience.

Sanitization Filtering

Inline 50-micron disks, automated chlorine dosing, and UV sterilizers eliminate waterborne viruses from reaching drinkers.

Biosecurity Infrastructure CAPEX

CAPEX: R 95,000

Constructs physical disinfection barriers to isolate clean livestock areas from vehicle and human vector contamination.

- **Perimeter Spray-Dip Boot:** Vehicle wheel wash sanitizes truck tires before site gate entry.
- **Shower-In Change Block:** Obligatory worker hygiene lock with farm-allocated uniforms.
- **Pod Footbaths:** Localized sanitizing steps outside every single nursery chamber threshold.

Functional Role

Eliminates pathogen introduction, preserving your zero-mortality target and reducing blanket veterinary medication costs across the production lifecycle.

IoT Sensors & Automation Systems

Ambient Environments

Continuous monitoring of temp, relative humidity, and ammonia gas levels inside the nursery pods.

Pulse Flow Meters

Inline flow tracking on pod water lines. Flags sudden consumption drops on the dashboard up to 48 hours before clinical illness signs appear.

Automated Valves

LoRaWAN triggers automated steam valve modulation to keep climate grids constant without manual oversight.

Contingency & Commissioning Details

CAPEX: R 180,000

Ensures expert engineering calibration during final structural assembly and system hand-over.

- **System Calibration:** Synchronizing solar output with BESS battery cycles and backup biogas loops.
- **Plumbing Sign-Off:** Flow rate validation across all sub-floor radiant steam grids.
- **Biological Validation:** Final chemical pressure testing and biosecurity audit clearance.

Functional Role

Provides third-party engineering compliance certification to secure capital drawdown approvals from international green energy funding syndicates.

Value Engineering Review

CAPEX Optimization

Standalone Nursery pods bypass double-roof interior construction costs, saving R150k in structural overheads.

Operational Savings

Gravity waste drainage and passive roof chimneys cut daily electrical base draws, preserving BESS lifespan.

Asset Defense

Decentralized pod units and water sterilization secure the zero-mortality target, stabilizing project cash flows.

INVESTMENT MODEL LOCKED

Total Project CAPEX: R 3,100,000

Lwandile Engineering Projects (Pty) Ltd

Mpumalanga JET Infrastructure | Precision Execution